Oblig2

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$$f(x) = e^{-x/4} \cdot \tan^{-1} x$$

$$u = e^{-x/4} \qquad v = \tan^{-1} x$$

$$u' = -\frac{1}{4} e^{-x/4} \qquad v' = \frac{1}{x^2+1}$$

$$f'(x) = -\frac{1}{4} e^{-x/4} \cdot \tan^{-1} x + e^{-x/4} \cdot \frac{1}{x^2+1}$$

$$f'(x) = e^{-x/4} \left(-\frac{1}{4} \tan^{-1} x + \frac{1}{x^2+1} \right)$$

$$-\frac{1}{4} \tan^{-1} x + \frac{1}{x^2+1} = 0 \qquad | \cdot 4$$

$$-\tan^{-1} x + \frac{4}{x^2+1} = 0 \qquad | \cdot -1$$

$$\tan^{-1} x - \frac{4}{x^2+1} = 0$$